UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 007 · Taylor's and Maclaurin's expansion · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

Contents

1. 1 · Why Taylor's and Maclaurin's series?
2. 2 · Taylor's and Maclaurin's series in one variable
3. 3 · Taylor's expansion of a function of two variables
4. 4 · Maclaurin's expansion of a function of two variables
5. Past-paper questions

1 Why Taylor's and Maclaurin's series?

Key points

1. Once a function is a polynomial, you can differentiate, integrate, find limits and study how it changes. ESA ★ T007 00:06:50
2. Applications (one variable): polynomials for complicated functions, definite integrals with no closed form, asymptotic behaviour, growth of functions, differential equations, the Lorentz factor, $E = mc^2$ at low speed, $\sin\theta \approx \theta$ for a pendulum, power-flow analysis, generating functions. ESA ★ S007·6 · S007·7 · T007 00:11:31 · T007 00:13:44
3. Applications (two variables): optimization (a linear or quadratic model), accurate distance measurement with the sine law, variance of complicated configurations, visualizing complicated functions. ESA ★ S007·12 · T007 00:21:22
4. A few terms are often enough: four terms of the sine series already follow $\sin x$ closely near 0, and ten terms overlap it (Figure G1.1). ESA ★ S007·8 · T007 00:15:24 · T007 00:16:10

5. Taylor's theorem is the basis of L'Hôpital's rule and the mean value theorems of Rolle, Lagrange and Cauchy; the series is a consequence of the theorem. ESA ★ TRANSCRIPT ONLY T007 00:03:33

Why?

Why does a calculator need series at all?
It can only add and multiply. A polynomial needs nothing else, so a function such as $e^{\tan x}$ is evaluated through polynomial approximations.

T007 00:01:02 · T007 00:01:35

Exam traps

Exam trap. Slide 007·3's limit $\lim \frac{\sin x - x}{x^4}$ does not exist; with x^3 the answer is $-\frac{1}{6}$ (W1.2).
ADDED FOR CLARITY S007·3

2 Taylor's and Maclaurin's series in one variable

Definitions

Taylor's series (one variable). If a function $f(x)$ has continuous derivatives of order $(n+1)$, then this function can be expressed as a polynomial at a point $x = a$ in the form $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \ldots$ S007·4 · T007 00:04:19

Maclaurin's series (one variable). If the function is expanded about the origin we get Maclaurin's series: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots$; it is a particular case of Taylor's series. S007·4 · T007 00:06:17

Key points

1. Keep only the first two terms of Taylor's series and you get $f(x) \approx f(a) + (x-a)f'(a)$: the tangent line, i.e. the derivative as a slope (Figure G2.1). ESA ★ T007 00:05:26 · T007 00:05:43
2. To get a standard series, differentiate repeatedly and put $x = 0$: every derivative of e^x is e^x, which is 1 at the origin. ESA ★ T007 00:09:10 · T007 00:09:44
3. Substitute into a known series rather than differentiating: $\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \ldots$. ESA ★★ T007 00:11:51
4. No Maclaurin series exists if f or a derivative is infinite at 0: $1/x$, $\log x$, $\sqrt{x}$. ESA ★ T007 00:10:18

Why?

When is Maclaurin's series valid?
f must have derivatives of every order at 0 (so none is infinite there), and the series must converge to f. e^x, $\sin x$, $\cos x$ converge for every x; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.
ADDED FOR CLARITY T007 00:10:01 · S007·5

Exam traps

Exam trap. The lecture says $1 + 2 + 2^2 + \ldots$ "gives -1". It diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ only for $|x| < 1$; $\frac{1}{1-2} = -1$ is outside that range. ADDED FOR CLARITY T007 00:07:52

Exam trap. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$ has plain denominators; slide 007·5 shows **3!** and **5!**, which belong to $\sin x$. ADDED FOR CLARITY S007·5

3 Taylor's expansion of a function of two variables

Definitions

Taylor polynomial of degree n. The terms of Taylor's series up to total degree n in $(x-a)$ and $(y-b)$; "expand up to second (third) degree terms" means stop there. Degree 1 is the tangent plane. ADDED FOR CLARITY S007·11 · T007 00:20:13

Statements in the class material

> *"Taylor's series of $f(x,y)$ about (a,b) is given by* $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x(a,b) + (y-b)f_y(a,b)\} + \frac{1}{2!}\{(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\} + \ldots$" S007·9 · T007 00:18:45

Key points

1. Lay out a table: f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} (and third order if asked), each in general and then its value at (a,b). Then substitute into E3.1. ESA ★★★ ADDED FOR CLARITY CW U1 P24 · ESA Jul-23 1b
2. There are four second-order partials, but $f_{xy} = f_{yx}$, so they combine into $2(x-a)(y-b)f_{xy}$. ESA ★★ T007 00:18:26
3. "In powers of $(x-1)$ and $(y-\frac{\pi}{4})$" means about the point $(1, \frac{\pi}{4})$. ESA ★★ ADDED FOR CLARITY ESA May-24 1b · CW U1 P27
4. Taylor's series gives approximate values: $\tan^{-1}\frac{0.9}{1.1}$ from the expansion of $\tan^{-1}\frac{y}{x}$ about $(1,1)$ with $h = 0.1$, $k = -0.1$. ESA ★★ CW U1 P26
5. The first-degree Taylor polynomial is the tangent plane; the second degree hugs the surface much better (slide 007·11, Figures G3.1–G3.2). ESA ★ S007·11 · T007 00:20:51

Why?

Why are the second-order coefficients 1, 2, 1?
Put $g(t) = f(a+th, b+tk)$ and use the one-variable Maclaurin series of g at $t=0$. By the chain rule $g'(0) = hf_x + kf_y$ and $g''(0) = h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}$ (the two equal mixed partials give the 2). Put $t = 1$.
ADDED FOR CLARITY T007 00:18:26

Exam traps

Exam trap. Evaluate every derivative at (a,b) before substituting; never leave $f_{xx}(x,y)$ in the answer. ADDED FOR CLARITY S007·9

Exam trap. Don't forget $\frac{1}{2!}$ and $\frac{1}{3!}$, and the middle coefficients **2** and **3**: the third-order bracket is $h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}$. ADDED FOR CLARITY ESA Dec-23 1b

Exam trap. Slide 007·11's polynomials ($1 + x + y$, $1 + x + xy + \frac{y^2}{2}$) are about $(1, 0)$, not the origin; about the origin they are $1 + x$ and $1 + x + xy$. ADDED FOR CLARITY S007·11 · T007 00:20:32

4 Maclaurin's expansion of a function of two variables

Definitions

Maclaurin's series (two variables). Taylor's series of $f(x, y)$ about the origin $(0, 0)$. S007·10 · T007 00:19:01

Statements in the class material

> *"Maclaurin's series for a function of two variables is given by* $f(x, y) = f(0, 0) + \frac{1}{1!}\{xf_x(0, 0) + yf_y(0, 0)\} + \frac{1}{2!}\{x^2 f_{xx}(0, 0) + 2xyf_{xy}(0, 0) + y^2 f_{yy}(0, 0)\} + \ldots$*"* S007·10

Key points

1. Memory aid from the lecture: f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. ESA ★★ T007 00:19:38
2. Shortcut and check: if f depends on one combination, use the one-variable series: $\sin(x + 2y) = (x + 2y) - \frac{(x+2y)^3}{6} + \ldots$, $\log(1 + x - y) = (x - y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3} - \ldots$ ESA ★★★ (added for clarity) CW U1 P25 · ESA Jun-24 1c · ESA Jan-24 s1 1b
3. None of f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} may be infinite at $(0, 0)$. ESA ★ T007 00:19:26

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A Taylor or Maclaurin expansion in two variables is asked in six of them, always as a 6–7 mark Q1 part. About a point (a, b): Dec 2023 Q1.b ($\sin x \sin y$, third degree), Jul 2023 Q1.b (the same function, second order) and May 2024 Q1.b ($e^x \cos y$ about $(1, \pi/4)$). About the origin: Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four of the six ask for third-degree terms, which the slides do not write out. Jan 2024 sets 3 and 4 have no Taylor question. One-variable series and the applications were never asked.

3 · Taylor's expansion of a function of two variables

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W3.1

Expand $f(x, y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W3.2

Expand $f(x, y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W3.3

Expand $f(x, y) = e^x \cos y$ in powers of $(x - 1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

4 · Maclaurin's expansion of a function of two variables

ESA Jan 2024 (7 marks) UE23MA141A set 1, Q1.b · solved in W4.1

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 (7 marks) UE23MA141A set 2, Q1.b · solved in W4.2

Expand $e^{ax} \sin by$ about origin up to third degree terms.

ESA June 2024 (6 marks) UE23MA141A Q1.c · solved in W4.3

Expand $f(x, y) = \sin(x + 2y)$ in Taylor's series up to third degree terms about the point $(0, 0)$.